

Título del documento (artículo)

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Resumen

Latex es una aplicación para preparar documentos aplicando un lenguaje de marcado. Permite producir textos escritos de alta calidad, especializado en composición de expresiones de matemáticas, física, fórmulas, desde una fórmula estándar para uso general hasta artículos científicos, libros, informes y tesis. El propósito de este ejemplo es que sirva como una plantilla de L^AT_EX para generar documentos útiles para hacer reportes de asignaturas científicas. Esto es, que la estructura de este documento replica la estructura del método científico a través de las secciones de: *Introducción*, *Métodos*, *Resultados*, *Discusión y Conclusiones* y soporta el uso de figuras, tablas, gráficas y referencias bibliográficas.

Palabras clave: keyword1, keyword2, keyword3, keyword4.

1. Introducción

La sección de introducción en un reporte científico es la parte inicial que presenta el problema de estudio, el estado del arte, la brecha en el conocimiento y el objetivo de la investigación, estableciendo el contexto y justificando su relevancia. Responde a las preguntas: ¿Qué?, ¿Por qué? y ¿Para qué?.

En LaTeX, una sección se emplea para comenzar una parte sustancial del documento. Proporciona un título descriptivo impreso, y, según la clase del documento, una numeración automática (como 1., 2.A o incluso "Sección

Práctica”), y suele añadir una entrada automática en la tabla de contenidos (ToC). También genera un salto de página o de columna si la clase de documento lo define así.

2. Métodos

La sección de métodos en un informe científico describe cómo se realizó el estudio o el experimento. Debe describir precisamente cómo se realizaron las observaciones, experimentos y análisis. La finalidad de esto es que se pueda replicar por otras personas y corroborar los resultados obtenidos.

Por ejemplo en la figura 1.



Figura 1: Descripción de la imagen. En este caso, un retrato de Confucio tomado de la Wikipedia.

2.1. Esto es el título de una subsección

Una subsección en LaTeX se define mediante el comando `subsection{}`. Este comando genera automáticamente un número de subsección y da formato al título fomentando la estructura lógica del documento. Sitúa el texto indicado en la clase del documento y permite hacer referencias mediante `label{}`.

Sigamos con las imágenes. Podemos hacer una tabla de imágenes utilizando el paquete `subcaption` o `subfig`. Por ejemplo, para insertar una imagen con 2x2 imágenes (es decir, 4 subfiguras). En la subfigura 2b, se ve a confucio, en la figura 2a, se puede ver a Lao-Tze, etc..



(a) Retrato de Lao-Tse (Wikipedia) (b) Retrato de Confucio (c) Retrato de Russell (d) Retrato de Sun-tzu

Figura 2: Una muestra de cuatro retratos

2.2. Formato de Texto

Para hacer listas:

- Texto en **Negritas**
- Texto en *Cursiva*
- Texto Máquina de escribir

Cuadro 1: Este es un ejemplo de un curso con calificaciones.

Alumno	Ex 1	Ex 2	Ex 3	Ex 4	Promedio
Ana Pérez	7.5	8.0	9.0	7.0	7.9
Juan López	6.0	7.0	8.5	6.5	7.0
María García	9.0	9.5	8.9	8.5	9.0
Carlos Ruiz	8.0	8.5	8.0	7.5	8.0

2.3. Tablas

Las tablas o cuadros no sirven para representar relaciones entre datos.

2.4. Ejemplo para mostrar código

Para mostrar código podemos usar el ambiente 'Verbatim'.

```
def fibonacci(n):
    fib_sequence = []
    a, b = 0, 1

    while len(fib_sequence) < n:
        fib_sequence.append(a)
        a, b = b, a + b
    return fib_sequence

n = int(input("Manual Sequence n value? \n"))
print("Fibonacci sequence:")
fibonacci_seq = fibonacci(n)
print(fibonacci_seq)
```

Esta palabra es large pero esta es más *Grande*

3. Ecuaciones y símbolos matemáticos

Se pueden poner en la misma línea con \$ ecuacion entre signo de peso\$ por ejemplo: $f(x) = x_{i,j}^{23} + xy + \epsilon$. Con dos signos de peso (\$\$) lo hace en línea separada.

$$f(x) = \log(x^n) + \exp^{\alpha_i x}$$

3.1. Ambiente para ecuaciones

También hay un ambiente para ecuaciones para poner referencias e índices.

La función de densidad de probabilidad de una distribución gaussiana (normal) con media μ y varianza σ^2 es:

$$\begin{aligned} A &= \frac{\pi r^2}{2} \\ &= \frac{1}{2} \pi r^2 \\ &= \frac{1}{2} \pi r^2 \\ &= \frac{1}{2} \pi r^2 \\ &= \frac{1}{2} \pi r^2 \\ &= \frac{1}{2} \pi r^2 \\ &= \frac{1}{2} \pi r^2 \\ &= \frac{1}{2} \pi r^2 \\ &= \frac{1}{2} \pi r^2 \\ &= \frac{1}{2} \pi r^2 \end{aligned} \tag{1}$$

Tell me, Muse, of that man, so ready at need, who wandered far and wide, after he had sacked the sacred citadel of Troy, and many were the men whose towns he saw and whose mind he learnt, yea, and many the woes he suffered in his heart on the deep, striving to win his own life and the return of his company. Nay, but even so he saved not his company, though he desired it sore. For through the blindness of their own hearts they perished, fools, who devoured the oxen of Helios Hyperion: but the god took from them their day of returning. Of these things, goddess, daughter of Zeus, whencesoever thou hast heard thereof, declare thou even unto us.

3.1.1. Subsubsección

Now all the rest, as many as fled from sheer destruction, were at home, and had escaped both war and sea, but Odysseus only, craving for his wife and

for his homeward path, the lady nymph Calypso held, that fair goddess, in her hollow caves, longing to have him for her lord. But when now the year had come in the courses of the seasons, wherein the gods had ordained that he should return home to Ithaca, not even there was he quit of labours, not even among his own; but all the gods had pity on him save Poseidon, who raged continually against godlike Odysseus, till he came to his own country. Howbeit Poseidon had now departed for the distant Ethiopians, the Ethiopians that are sundered in twain, the uttermost of men, abiding some where Hyperion sinks and some where he rises.

Párrafo There he looked to receive his hecatomb of bulls and rams, there he made merry sitting at the feast, but the other gods were gathered in the halls of Olympian Zeus. Then among them the father of men and gods began to speak, for he bethought him in his heart of noble Aegisthus, whom the son of Agamemnon, far-famed Orestes, slew.

Subpárrafo Thinking upon him he spake out among the Immortals: ‘Lo you now, how vainly mortal men do blame the gods! For of us they say comes evil, whereas they even of themselves, through the blindness of their own hearts, have sorrows beyond that which is ordained. Even as of late Aegisthus, beyond that which was ordained, took to him the wedded wife of the son of Atreus, and killed her lord on his return, and that with sheer doom before his eyes, since we had warned him by the embassy of Hermes the keen-sighted, the slayer of Argos, that he should neither kill the man, nor woo his wife. For the son of Atreus shall be avenged at the hand of Orestes, so soon as he shall come to man’s estate and long for his own country. So spake Hermes, yet he prevailed not on the heart of Aegisthus, for all his good will; but now hath he paid one price for all.’

And the goddess, grey-eyed Athene, answered him, saying: ‘O father, our father Cronides, throned in the highest; that man assuredly lies in a death that is his due; so perish likewise all who work such deeds! But my heart is rent for wise Odysseus, the hapless one, who far from his friends this long while suffereth affliction in a sea-girt isle, where is the navel of the sea, a woodland isle, and therein a goddess hath her habitation, the daughter of the wizard Atlas, who knows the depths of every sea, and himself upholds the tall pillars which keep earth and sky asunder. His daughter it is that holds

the hapless man in sorrow: and ever with soft and guileful tales she is wooing him to forgetfulness of Ithaca. But Odysseus yearning to see if it were but the smoke leap upwards from his own land, hath a desire to die. As for thee, thine heart regardeth it not at all, Olympian! What! Did not Odysseus by the ships of the Argives make thee free offering of sacrifice in the wide Trojan land? Wherefore wast thou then so wroth with him, O Zeus?’

The “Odyssey” (as every one knows) abounds in passages borrowed from the “Iliad”; I had wished to print these in a slightly different type, with marginal references to the “Iliad,” and had marked them to this end in my MS. I found, however, that the translation would be thus hopelessly scholasticised, and abandoned my intention. I would nevertheless urge on those who have the management of our University presses, that they would render a great service to students if they would publish a Greek text of the “Odyssey” with the Iliadic passages printed in a different type, and with marginal references. I have given the British Museum a copy of the “Odyssey” with the Iliadic passages underlined and referred to in MS.; I have also given an “Iliad” marked with all the Odyssean passages, and their references; but copies of both the “Iliad” and “Odyssey” so marked ought to be within easy reach of all students.

Any one who at the present day discusses the questions that have arisen round the “Iliad” since Wolf’s time, without keeping it well before his reader’s mind that the “Odyssey” was demonstrably written from one single neighbourhood, and hence (even though nothing else pointed to this conclusion) presumably by one person only—that it was written certainly before 750, and in all probability before 1000 B.C.—that the writer of this very early poem was demonstrably familiar with the “Iliad” as we now have it, borrowing as freely from those books whose genuineness has been most impugned, as from those which are admitted to be by Homer—any one who fails to keep these points before his readers, is hardly dealing equitably by them. Any one on the other hand, who will mark his “Iliad” and his “Odyssey” from the copies in the British Museum above referred to, and who will draw the only inference that common sense can draw from the presence of so many identical passages in both poems, will, I believe, find no difficulty in assigning their proper value to a large number of books here and on the Continent that at present enjoy considerable reputations. Furthermore, and this perhaps is an advantage better worth securing, he will find that many puzzles of the “Odyssey” cease to puzzle him on the discovery that they arise from over-saturation with the “Iliad.”

Other difficulties will also disappear as soon as the development of the poem in the writer's mind is understood. I have dealt with this at some length in pp. 251-261 of "The Authoress of the Odyssey". Briefly, the "Odyssey" consists of two distinct poems: (1) The Return of Ulysses, which alone the Muse is asked to sing in the opening lines of the poem. This poem includes the Phaeacian episode, and the account of Ulysses' adventures as told by himself in Books ix.-xii. It consists of lines 1-79 (roughly) of Book i., of line 28 of Book v., and thence without intermission to the middle of line 187 of Book xiii., at which point the original scheme was abandoned.

(2) The story of Penelope and the suitors, with the episode of Telemachus' voyage to Pylos. This poem begins with line 80 (roughly) of Book i., is continued to the end of Book iv., and not resumed till Ulysses wakes in the middle of line 187, Book xiii., from whence it continues to the end of Book xxiv.

In "The Authoress of the Odyssey", I wrote:

the introduction of lines xi., 115-137 and of line ix., 535, with the writing a new council of the gods at the beginning of Book v., to take the place of the one that was removed to Book i., 1-79, were the only things that were done to give even a semblance of unity to the old scheme and the new, and to conceal the fact that the Muse, after being asked to sing of one subject, spend two-thirds of her time in singing a very different one, with a climax for which no-one has asked her. For roughly the Return occupies eight Books, and Penelope and the Suitors sixteen.

4. Ambientes

En LaTeX, los ambientes son bloques definidos por `begin...` y que estructuran y formatean contenido. Ejemplos: 'itemize' (listas con viñetas), 'equation' (ecuaciones), 'figure' (imágenes).